

FSOC Tracker: User Manual

AI-based virtual camera tracking system for coarse alignment of mobile Free Space Optical Communication (FSOC) terminals. Smart India Hackathon problem statement SIH26169, Department of Space / ISRO Space Applications Centre.

1. What the application does

The application simulates the coarse alignment stage of an FSOC link entirely in software.

- It draws a scene (the "screen", at least 2000 x 2000 pixels) with a moving optical beacon.
- A virtual pan-tilt camera looks at a 640 x 480 pixel window of that scene (4 x 3 degrees) and can turn at a limited speed, like a real motorised mount.
- Disturbances can be added: fog, haze, rain, low light, atmospheric turbulence, platform motion, camera vibration, and sensor noise.
- The tracker detects the beacon, measures its centre to a fraction of a pixel, predicts where it is going, and steers the camera window to keep it centred.
- Every run writes a per-frame log, a summary and an automatic performance report into one folder named `FSOC_<sim|video>_<name>_seed<N>_<date-time>` (section 7).
- For Benchmark 2, an `.mp4` video can be opened in place of the simulated scene.

2. Installation

2.1 Standalone executable (no Python needed)

Archives are built for every desktop platform in use:

Platform	Archive	Run
Windows 10 or 11, 64-bit	<code>FSOC-Tracker-windows-x64.zip</code>	<code>FSOC-Tracker.exe</code> (window); <code>FSOC-Tracker-cli.exe</code> for the command line
Linux, 64-bit (Ubuntu 22.04 or newer, Debian 12, BOSS, RHEL 9, Fedora)	<code>FSOC-Tracker-linux-x64.tar.gz</code>	<code>./FSOC-Tracker</code>
macOS 13 or newer, Intel	<code>FSOC-Tracker-macos-intel.zip</code>	<code>FSOC-Tracker</code>
macOS 13 or newer, Apple silicon	<code>FSOC-Tracker-macos-arm64.zip</code>	<code>FSOC-Tracker</code>

A demonstration video (`FSOC-Tracker-demo.mp4` , about four minutes, composed from the engine's own frames) and a 10 to 15 minute live demo script (`docs/DEMO_SCRIPT.md`) accompany the builds.

1. Download the archive for your platform from the downloads page.
2. Extract it anywhere the user can write to (Desktop, Documents, home folder). Do not run it from inside the archive.

3. Run the executable named above. The first start takes a few seconds.

Platform notes:

- Windows: SmartScreen may show "Windows protected your PC" because the executable is not code-signed. Choose More info, then Run anyway. No administrator rights are needed.
- Linux: the archive carries Qt and all Python libraries. The system needs the usual X11 or Wayland client libraries, present on every desktop install. On a minimal server image install `libxcb-cursor0 libxkbcommon-x11-0 libegl1` (Debian and Ubuntu names). If the file is not executable after extraction, run `chmod +x FSOC-Tracker`.
- macOS: Gatekeeper may say the application cannot be verified. Open System Settings, Privacy and Security, and choose Open Anyway, or run `xattr -dr com.apple.quarantine FSOC-Tracker` on the extracted folder once.

Results are written to a `results` folder next to the executable. Every archive is produced by the same build workflow, which also runs the test suite and a smoke test of the packaged executable on that platform before publishing it.

2.3 Web application

The same program is also served as a web application, so no installation at all is needed on any platform: the project site opens it directly in a current browser (Chrome, Edge, Firefox or Safari, on Windows, Linux, macOS, or a tablet). The web page and the desktop window are the same interface: the same toolbar (Start, Pause, Step, Stop, Speed, Open video, Simulator, Save scenario, Screenshot, Results, Manual, About), the same parameter panel with every field, the same six tiles, views, four plots, telemetry column and end-of-run summary, and the same keyboard keys. Both are generated from one definition in the code (`fsoc_tracker/ui_shared.py`) and run the same engine, so a given scenario and seed give the same numbers and the same report.

What differs follows from running on a server, not from the program:

- The views in the browser are sent as snapshots about 15 times a second; the log, the plots and the report still cover every frame.
- One run executes at a time per server; a second visitor watches the run in progress.
- Files are downloaded from the browser instead of opened in a folder (Results lists them).
- The page also offers the desktop application for download (the Desktop app button).

2.2 From source

Requires Python 3.11 or 3.12.

```
git clone <repository>
cd SIH169
python -m venv .venv
.venv/bin/pip install -e ".[dev]"          # Windows: .venv\Scripts\pip install -e ".
[dev]"
.venv/bin/fsoc-tracker-gui                # start the desktop application
```

Optional, to retrain the AI detector: `pip install -e ".[train]"` then python `training/train_heatmap.py`.

3. The main window

```
+-----+-----+-----+
--+
| Parameters      | State | Acquisition | Track err | Centroid | Lock | FPS |
Telemetry |
| (scrollable,    +-----+ one value per
| hover for help) | Screen view          | Camera view          | line, every
|                  | whole scene, camera  | 640 x 480 with HUD   | frame
|                  | window, trails, 2 deg | capture ring, error |
|                  | grid, legend         | vector, 1 deg bar   |
|                  +-----+
|                  | Errors      | Gimbal command      |
|                  | Timing      | Tracker state        |
+-----+-----+-----+
--+
| Status: what is running and where results go          [=====]
|
+-----+-----+-----+
+
```

Toolbar, left to right

Control	Action
Scenario	Pick a ready-made test case from <code>configs/scenarios</code> . "Custom" keeps whatever is in the panel.
Start / Pause / Step / Stop	Run control. Space starts or pauses, N steps one frame while paused, Esc stops.
Speed	0.25x to 4x real time, or Max speed (shows the true processing rate).
Open video (Benchmark 2)	Choose an <code>.mp4</code> ; the simulator is bypassed and the video frames become the scene. Ctrl+O.
Use simulator	Return to the simulated scene.
Save scenario	Save the panel as a <code>.yaml</code> ; it appears in the Scenario list. Ctrl+S.
Screenshot	Save a PNG of the window into <code>results/</code> . Ctrl+P.
Results folder	Open <code>results/</code> .

Control	Action
User manual	Open this manual.
Help	Legend and output summary.

Tiles (live specification check)

Six tiles above the pictures update every frame and turn green when the problem statement specification is met, amber when it is not: tracker state, acquisition time (2 s or less), mean tracking error since acquisition (10 px or less), mean centroiding error, lock retention (target loss under 5 percent), and processing FPS (20 or more). In video mode the two error tiles read "n/a" because the video carries no ground truth.

Screen view (left picture)

The whole scene, downscaled, with a 2 degree grid. Cyan rectangle: the camera window and its centre. Orange circle: the true beacon (simulator only). Green cross: the tracker's estimate. Grey circles: other targets. Every target carries its name; the designated one is marked "(designated)" in the signal colour, the others are muted. Before a run the view shows a preview of the scene at $t = 0$ with every target named; click a target to make it the designated one. Faint trails: where the beacon and the window have been. The legend is printed along the bottom edge.

Camera view (right picture)

What the camera window sees, contrast-stretched for display. Cyan cross: boresight. Dashed ring: the capture radius, green when locked. Line from the centre: the pointing error vector. Green box: this frame's detection. Orange dot: the prediction, with a fainter ring for the estimator's uncertainty. Green trail: recent estimates. A 1 degree scale bar sits bottom right. The banner shows the state, tracking error in pixels and arcseconds, confidence and which detector produced the measurement; the bottom line shows pan, tilt, the commanded rates, saturation and processing time.

Plots

- Errors: tracking error (window centre to beacon, cyan) and centroiding error (measured centre to true centre, orange). Dashed line: the 10 pixel specification.
- Gimbal command: pan and tilt rate commands with the rate limit dashed.
- Processing time per frame with the 20 FPS budget (50 ms) dashed.
- Tracker state: SEARCH, VERIFY, TRACK, COAST, REACQUIRE.

Telemetry column

Every internal quantity for the current frame: state, detector, candidates, confidence, SNR, pose, commands, detection, estimate, velocity, uncertainty, motion model probabilities, truth, errors and timing. Before a run it shows a short how-to.

4. Parameters

Every row of the problem statement parameter table has a control. Values outside the suggested range are allowed where the table says "user-defined". Every numeric input is clamped to its accepted range in the engine, for both applications, and the scenario check (section 4.1) says what a value means.

Run

- Name, Seed: the seed makes a run exactly repeatable.
- Duration (s): length of a simulator run. Ignored for video input (the whole video runs).
- Extra targets: number of additional beacons (decoys) added with random paths.
- Identical look: the generated extra targets copy target 1's shape, size and brightness. Only a start or click cue can then tell the designated one apart.
- Designated: the target the tracker must follow (PS: "a designated moving target"). The report scores this one. The list shows the target names.
- Designation: how the tracker is told which one it is. `appearance` : by its configured shape, size and brightness. `start` : also told where it starts, as an operator or GPS cue would. `cue` : a point you give.
- Cue (x,y): the point for designation `cue` , in screen pixels. Clicking the scene sets it; you can also type it. While searching, the tracker takes the strong candidate nearest the cue; after a loss its last estimate becomes the cue.
- Edit target: which target the Target section below shows and edits.

Screen (rows 1 to 2)

- Width, Height: scene size in pixels. Default 2000 x 2000.
- Colour camera: renders a three-channel colour frame (PS row 2, optional); the tracker always works on luminance, so results are identical.
- Background: starfield, terrain, gradient or flat. Sky level and star density.

Camera (rows 3 to 6, 13 to 15)

- Width, Height: camera window in pixels. Default 640 x 480.
- FOV width, height: degrees. Default 4 x 3. Degrees per pixel is derived from these.
- Update rate: frames per second. Default 30.
- Max pan, Max tilt: degrees per second. Default 5.
- Max accel, Command latency: gimbal realism.
- Hard mode: the tracker sees only the pixels inside the window and must search.

Target (rows 7 to 12)

Shows the target chosen in Edit target. - Name: shown on the views, in the telemetry and in the report. Empty means "Target N". - Shape: square (default), circle, gaussian, cross, ring, diamond, custom. Width and Height in pixels, set separately (default 10 x 10; the PS range is 5-20 x 5-20). A square with unequal sides is a rectangle, a circle an ellipse. Peak intensity. - Custom shape (0/1 rows): for shape `custom`, rows of 0 and 1 separated by `;`, stretched to width x height. `010;111;010` is a plus. A PNG path also works. - Motion: line, circular, figure8, random, spiral, sinusoidal, waypoints, static. For `waypoints`, the Waypoints field takes screen-pixel points as `x,y; x,y; ...`; the beacon follows

them at Speed and loops (the PS row 12 user-defined path). - Speed, Radius, Period, Heading: path parameters. Start: random, centre, or a typed `x,y` in screen pixels (for example `400,1500`). - Blink: optional intensity modulation in Hz. 0 is steady.

Disturbances (rows 21 to 25)

- Salt and pepper in percent of pixels (10 = 10 percent; accepted 0 to 50; scenario files store it as a fraction, 0.10), Gaussian sigma (up to 20), Poisson.
- Camera jitter: pixels per frame, up to 20.
- Atmosphere preset: clear, haze, fog, rain, lowlight. Selecting a preset fills contrast, brightness, blur and turbulence; each can then be edited.
- Platform motion: none, linear, circular, random, spiral, figure8; speed in pixels per frame (up to 20) and period.

Tracker

- Detector: hybrid (classical first, AI fills gaps), classical, cnn.
- Use picture-shift estimate: phase correlation as a vibration hint.
- Controller gains, deadband, capture radius, estimator lag, minimum confidence to acquire.
- Faint path: when nothing reaches the acquisition confidence, weak detections (threshold `faint_threshold_k`, default 3 sigma) are linked across frames and a motion-consistent chain with mean SNR above `faint_snr_min` (default 3.5) is promoted. Used automatically for dim beacons in low light; a static dim beacon is not covered by this path.
- Hard mode (Camera section) restricts the tracker to the window; SEARCH then flies an expanding square spiral of window-sized cells at the rate limit. A full sweep of a 2000 px screen at 5 deg/s takes about 12 s, so acquisition in hard mode is 3 to 12 s depending on where the beacon is.

4.1 Scenario check

Below the Run section, both applications show notes on the current values. They also appear in the status bar at Start, in the end-of-run dialog, on page 1 of the report and in the summary. Values inside the PS envelope give no note.

Note	Meaning
Corrected	A value was outside the accepted range and was clamped, for example salt and pepper 13 (a fraction) becomes 0.5.
Beyond the PS	A value is outside the PS table, and the row is named: screen below 2000 x 2000 (row 1), update rate below 30 Hz (row 5) or 20 Hz (row 15), pan or tilt outside 5 to 10 deg/s (rows 13, 14), size outside 5-20 x 5-20 (row 10), custom shape without a mask, salt and pepper above about 10% (row 21), Gaussian sigma above 20 (row 22), jitter above 20 px/frame (row 23), platform above 20 px/frame (row 25), turbulence above 0.6, contrast below 0.4. The PS targets are not promised for it.
Cannot be met	A physical limit: the designated beacon moves faster than the camera turns (800 px/s at 5 deg/s and the default FOV), or above 70 percent of it; jitter plus platform motion above the camera's turn per frame (26.7 px/frame at the defaults); salt and pepper at 50%; designation <code>cue</code> without a point; designation <code>start</code> with a video; other targets that look the same as the designated one in appearance mode.

5. Running Benchmark 2 (video input)

1. Click "Open video (Benchmark 2)" and choose the `.mp4` file. The application reads the file's real facts (displayed size with any rotation tag applied, average frame rate with a variable-rate warning, exact frame count, length) and calibrates the settings to them: the screen becomes the video's size and the update rate its frame rate. Disturbance settings and the target's motion fields are locked, because the video already contains them. The target's appearance (name, shape, width, height, mask, intensity) stays editable: for a video it is the statement of what to look for. Set it to the beacon in the video. Only the designated target's look is used. To point the tracker at one beacon, click it on the first frame: this sets a designation cue and Designation switches to `cue`.
2. Set the camera window size, FOV and rate limits if the graders specify them; these are not in the file. Degree readouts depend on the FOV you set.
3. Click Start. The video frames are used as the scene; nothing is drawn by the simulator.
4. When the run ends, `frames.csv` contains the measured beacon centre for every frame (`det_x`, `det_y`) and `report.pdf` contains acquisition time, re-acquisition time, lock retention rate and FPS. Tracking and centroiding error against truth are not available because the video carries no ground truth.

From the command line: `fsoc-tracker video path/to/file.mp4`

6. Scenario files and the command line

A scenario is a YAML file with the same fields as the parameter panel. Example:

```
name: fog_circular
seed: 6
duration_s: 30
targets:
  - {name: Remote terminal, shape: square, size_px: 10, height_px: 10, motion:
circular, radius_px: 400, period_s: 15}
disturbance:
  atmosphere: fog
  gaussian_sigma: 8
```

Target fields: `name`, `shape`, `size_px` (width), `height_px` (0 = same as the width), `mask` (for custom), `intensity`, `motion` and its path fields, `start` (`random`, `centre` or `"x,y"`). Run fields for designation: `designated` (index of the target to follow, default 0), `designation` (`appearance`, `start` or `cue`) and `designation_cue` (`"x,y"` in screen pixels). `disturbance.salt_pepper_frac` is a fraction (0.10 = 10 percent).

Commands:

```
fsoc-tracker run --scenario configs/scenarios/fog_circular.yaml [--seed 3] [--
duration 20]
fsoc-tracker video path/to/file.mp4
```

```
fsoc-tracker batch --scenario configs/scenarios/*.yaml --seeds 0-49
fsoc-tracker gui
```

`batch` runs every scenario over every seed and writes `envelope.md`, a table of the worst and mean results per scenario.

Evaluator-supplied files

Benchmark Performance 1 gives you scenario parameters. Copy `configs/scenarios/TEMPLATE_evaluator.yaml`, which lists every field with the PS row it implements, fill in the given values, save it under `configs/scenarios/`, and it appears in the Scenario picker of both applications; or run it with `fsoc-tracker run -s <file>`. Fields the evaluators do not specify stay at the PS defaults.

Benchmark Performance 2 gives you `.mp4` files. No scenario file is needed: open the file in the desktop application, drop it on the web app, or run `fsoc-tracker video <file>`. The per-frame CSV then carries `det_x`, `det_y` (the measured centroids) for comparison with the evaluators' predefined values, and the report carries acquisition, re-acquisition, lock retention and FPS.

7. Output files

Every run gets one label and one folder named after it:

<code>results/FSOC_sim_<scenario>_seed<N>_<YYYYMMDD-HHMMSS>/</code>	simulated run
<code>results/FSOC_video_<file name>_<YYYYMMDD-HHMMSS>/</code>	Benchmark 2 run on a video

for example `results/FSOC_sim_clear_line_seed7_20260922-101530/`. The four files inside carry the same label as a prefix, so a report copied out of its folder still says which scenario, seed and time it belongs to:

File	Contents
<code><label>_frames.csv</code>	One row per frame: time, state, detection, estimate, camera pose, commands, truth, errors, processing time.
<code><label>_summary.json</code>	All metrics with their definitions and pass/fail against the specification; <code>designation</code> (followed target, index, mode, cue, all target names, ambiguous frames, redesignations) and <code>checks</code> (the scenario check notes).
<code><label>_report.pdf</code>	Specification check, the Targets and Followed lines, the scenario check notes and the metric table (continued on the next page when long); time series; paths, histograms and model probabilities.
<code><label>_scenario.yaml</code>	The exact parameters used, so the run can be repeated.

With `--out <folder>` on the command line the folder is yours; the files inside are still labelled. Batch runs write `results/batch/<time>/<scenario>_seed<N>/` with labelled files and an `envelope.md` table. The web app names its downloads the same way.

8. Metric definitions

Metric	Definition
Acquisition time	First frame in TRACK with the tracked beacon within the capture radius (30 px) of the window centre. Spec: 2 s or less.
Tracking error	Distance from the true beacon to the window centre, over frames after acquisition. Spec: 10 px or less (mean).
Tracking error, vibration removed	The same with the per-frame camera vibration subtracted from the truth: the part a rate-limited gimbal can physically follow.
Centroiding error	Distance from the measured beacon centre to the true centre.
Tracked	Percentage of frames after acquisition in which the tracker held the beacon (state TRACK), whether or not the camera had it centred. A high tracked rate with a low lock rate means the camera, not the tracker, could not keep up.
Lock retention	Percentage of frames after acquisition in TRACK with the beacon inside the capture radius. Target loss is 100 minus this. Spec: loss under 5 percent.
Re-acquisition time	Time from losing lock to regaining it. Spec: 1 s or less.
FPS	1 divided by per-frame processing time, averaged. Spec: 20 or more.

9. Troubleshooting

- A value changed after you typed it: it was outside the accepted range and was clamped. The scenario check shows it as "Corrected" with the old and new value.
- The beacon is never acquired: read the scenario check. "Cannot be met" notes name the setting that makes the run impossible.
- The window does not start on Linux: install `libxcb-cursor0` (Qt 6 requirement).
- Low frame rate: reduce the screen size, disable Poisson noise, or uncheck Real-time pacing to see the true processing speed.
- "Cannot open video": the file must be readable by OpenCV (H.264 `.mp4` is safest).
- The AI detector is not used: `models/beacon_heatmap.onnx` is missing; the classical detector runs alone. Retrain with `python training/train_heatmap.py`.